

Frontend Web Development Projects *HTML5, CSS3, JavaScript, Responsive Design, Vercel*

- Designed and developed multiple responsive business websites, including a cafe, hotel, salon, and creative studio, following modern UI/UX principles.
- Built reusable components, interactive user interfaces, mobile-first layouts, and optimized cross-browser compatibility for improved user experience.
- Deployed production-ready websites on Vercel while ensuring accessibility, performance optimization, and responsive design across all devices.

Spiking U-Net for Speech Enhancement *Python, PyTorch, Signal Processing*

- Designed and implemented a Spiking Neural Network (SNN)-based U-Net architecture for speech enhancement and background noise suppression.
- Developed the complete machine learning pipeline including audio preprocessing, model training, evaluation, and visualization of experimental results.
- Improved speech intelligibility using deep learning techniques for real-world noisy audio environments.

CERTIFICATIONS

GenAI Powered Data Analytics Job Simulation | *Tata Forage*

Cisco Networking Academy | *Python Essentials, Modern AI*

React (Basic) | *HackerRank*

ACHIEVEMENTS

Patent Application Filed *2025*

- Contributed to the development of an offline AI-powered companion system, leading to the filing of a patent application.

1st Place – **TECHSYNAPSE TECHBRIDGE National Hackathon** *Mar 2026*

- Secured 1st place by developing an FPGA-based VLSI solution during a 24-hour national hackathon.

Top 100 Team – **Smart India Hackathon (SIH) Internal Selection** *Sep 2025*

- Project **MITRA** was shortlisted among the Top 100 teams during the university-level Smart India Hackathon selection.

3rd Prize – **TEJAS 2K25 Engineering Expo** *Jan 2025*

- Awarded 3rd place for presenting **MITRA**, an offline AI-powered companion system.

TECHNICAL SKILLS

Programming: Java, Python, C, JavaScript

Frontend: React.js, HTML5, CSS3, Responsive Web Design

Backend: Spring Boot, REST APIs

Databases: MySQL

AI Technologies: Gemini API, OCR, Prompt Engineering, Local Language Models

Tools: Git, GitHub, VS Code, Vercel, Render

Concepts: API Integration, Component-Based Architecture, Mobile-First Design, UI/UX Design

LEADERSHIP & ACTIVITIES

IEEE CAS Student Chapter *PCB Design Workshop*

- Coordinated and facilitated a hands-on PCB Design Workshop, guiding participants through electronic circuit design, PCB layout, and hardware development fundamentals.

NodeMCU IoT Workshop *Workshop Organizer*

- Organized and managed a two-day NodeMCU IoT Workshop, introducing participants to embedded systems, IoT concepts, and practical implementation.

Hacktronics 2.0 National-Level Hackathon *Volunteer*

- Contributed to event operations, participant coordination, and the successful execution of technical sessions during the national-level hackathon.